

Practical 5

Roll No. :S006 Name:Sejal Manohar Deshmukh Class :SYIT Batch : 01 Date of Assignment :12/08/2026 Date/Time of Submission :12/08/2026

```
import pandas as pds
```

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

```
import pandas as pd

data = {
    "name": ["Siya", "Abhay", "Vikram"],
    "age": [22, 24, 20],
    "qualified": [True, False, False]
}
df = pd.DataFrame(data)
print(df)
```

	name	age	qualified
0	Siya	22	True
1	Abhay	24	False
2	Vikram	20	False

5b. Write a python code to show statistical information on given data set.

```
import statistics
```

```
print(statistics.median([1, 3, 5, 7, 9, 11, 13]))
print(statistics.median([1, 3, 5, 7, 9, 11]))
print(statistics.median([-11, 5.5, -3.4, 7.1, -9, 22]))
```

```
7
6.0
1.05
```

```
print(statistics.mean([1, 3, 5, 7, 9, 11, 13]))
print(statistics.mean([1, 3, 5, 7, 9, 11]))
print(statistics.mean([-11, 5.5, -3.4, 7.1, -9, 22]))
```

```
7
6
1.8666666666666667
```

```
print(statistics.mode([1, 3, 3, 3, 5, 7, 79, 11]))
print(statistics.mode([1, 1, 3, -5, 7, -9, 11]))
print(statistics.mode(['red', 'green', 'blue', 'red']))
```

```
3
1
red
```

```
print(statistics.stdev([1, 3, 5, 7, 9, 11]))
print(statistics.stdev([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.stdev([-11, 5.5, -3.4, 7.1]))
print(statistics.stdev([1, 30, 50, 100]))
```

```
3.7416573867739413
0.6925797186365383
8.414471660973927
41.67633221226007
```

```
print(statistics.variance([1, 3, 5, 7, 9, 11]))
print(statistics.variance([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.variance([-11, 5.5, -3.4, 7.1]))
print(statistics.variance([1, 30, 50, 100]))
```

```
14
0.47966666666666663
70.80333333333333
1736.9166666666667
```

5c. Write a python code to create pandas series from dictionaries.

```
dictionary = {'Siya': 5100, 'Abhay': 10000, 'Vikram': 830}
series = pd.Series(dictionary, index=['Vikram', 'Siya', 'Abhay'])

print(series)
```

```
Vikram      830
Siya       5100
Abhay     10000
dtype: int64
```

```
dictionary = {'Arjun': 10, 'Ram': 20, 'Krishana': 30}
series = pd.Series(dictionary)

print(series)
```

```
Arjun      10
Ram        20
Krishana   30
dtype: int64
```

Q5. Write a python code to demonstrate filter pandas series with Boolean arrays.

```
data = {
    "name": ["Siya", "Vikram", "Abhay"],
    "age": [5000, 4000, 3000],
    "qualified": [True, False, False]
}
df = pd.DataFrame(data)

newdf = df.filter(items=["name", "age"])
print(newdf)
```

```
   name  age
0  Siya  5000
1  Vikram 4000
2  Abhay  3000
```